==========

CAPSTONE PROJECT

NOTEBOOK 29: FINAL STRUCTURED MULTI-LABEL UPGRADE

==========

Purpose:

This notebook performs one final structured improvement round

for the candidate-150 multi-label genre task.

The goal is to:

1. Train 3 lightweight structured multi-label models

2. Compare them on validation performance

3. Tune per-label thresholds for the best model

4. Add a label-cap decoding rule to reduce over-prediction

5. Save the final improved structured baseline

==========

In [1]:

```
# ============================================================
# 1. IMPORT LIBRARIES
# ============================================================

import os
import joblib
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.multiclass import OneVsRestClassifier
from sklearn.linear_model import SGDClassifier, PassiveAggressiveClassifier
from sklearn.metrics import (
    f1_score,
    hamming_loss,
    accuracy_score,
    precision_score,
    recall_score
)
```

In [2]:

```
# ============================================================
# 2. LOAD MULTI-LABEL MASTER TABLE AND FEATURES
# ============================================================

master_df = pd.read_csv("../data/processed/multilabel_candidate_master_table.csv")

features = pd.read_csv(
    "../data/raw/metadata/features.csv",
    header=[0, 1, 2],
    index_col=0
)

features.index = features.index.astype(int)

print("Candidate multi-label master shape:", master_df.shape)
```

```
print("Features shape:", features.shape)

display(master_df.head())
```

```
Candidate multi-label master shape: (81574, 157)
Features shape: (106574, 518)
```

	track_id	split	subset	genre_top	title	audio_path	a
0	20	training	large	NaN	Spiritual Level	../data/raw/audio/fma_large\000\000020.mp3	
1	26	training	large	NaN	Where is your Love?	../data/raw/audio/fma_large\000\000026.mp3	
2	30	training	large	NaN	Too Happy	../data/raw/audio/fma_large\000\000030.mp3	
3	46	training	large	NaN	Yosemite	../data/raw/audio/fma_large\000\000046.mp3	
4	48	training	large	NaN	Light of Light	../data/raw/audio/fma_large\000\000048.mp3	

5 rows × 157 columns

In [3]:

```
# ============================================================
# 3. IDENTIFY LABEL COLUMNS
# ============================================================

label_cols = [
    col for col in master_df.columns
    if col.startswith("genre_") and col.replace("genre_", "").isdigit()
]

print("Number of candidate label columns:", len(label_cols))
print("First 10 label columns:", label_cols[:10])

print("\nNon-label genre-like columns:")
print([col for col in master_df.columns if col.startswith("genre_") and col not in
label_cols])
```

```
Number of candidate label columns: 150
First 10 label columns: ['genre_1', 'genre_2', 'genre_3', 'genre_4', 'genre_5',
'genre_6', 'genre_7', 'genre_8', 'genre_9', 'genre_10']

Non-label genre-like columns:
['genre_top']
```

In [4]:

```
# ============================================================
# 4. REMOVE TRACKS WITH NO CANDIDATE LABELS
# ============================================================

master_df["candidate_label_count"] = master_df[label_cols].sum(axis=1)

usable_df = master_df[master_df["candidate_label_count"] >
0].copy().reset_index(drop=True)

print("Original candidate master shape:", master_df.shape)
print("Usable candidate multi-label shape:", usable_df.shape)

print("\nSplit distribution after filtering:")
print(usable_df["split"].value_counts())

print("\nCandidate label count summary:")
print(usable_df["candidate_label_count"].describe())
```

```
Original candidate master shape: (81574, 158)
Usable candidate multi-label shape: (79343, 158)

Split distribution after filtering:
split
training      62452
test           8548
validation     8343
Name: count, dtype: int64

Candidate label count summary:
count    79343.000000
mean         3.679329
std          1.735141
min          1.000000
25%          3.000000
50%          4.000000
75%          5.000000
max         31.000000
Name: candidate_label_count, dtype: float64
```

In [5]:

```
# ============================================================
# 5. ALIGN FEATURES TO USABLE TRACKS
# ============================================================

usable_df["track_id"] = usable_df["track_id"].astype(int)
track_ids = usable_df["track_id"].tolist()

features_aligned = features.reindex(track_ids).copy()

missing_feature_rows = features_aligned.isna().all(axis=1)
num_missing = int(missing_feature_rows.sum())

print("Tracks requested:", len(track_ids))
print("Rows missing from features:", num_missing)
```

```
if num_missing > 0:
    usable_df = usable_df.loc[~missing_feature_rows].reset_index(drop=True)
    features_aligned = features_aligned.loc[~missing_feature_rows].copy()

features_aligned = features_aligned.reset_index(drop=True)

print("Aligned features shape:", features_aligned.shape)
print("Usable dataframe shape:", usable_df.shape)
```

```
Tracks requested: 79343
Rows missing from features: 0
Aligned features shape: (79343, 518)
Usable dataframe shape: (79343, 158)
```

In [6]:

```
# ============================================================
# 6. FLATTEN FEATURE COLUMNS
# ============================================================

features_aligned.columns = [
    "_".join([str(level) for level in col]).strip()
    for col in features_aligned.columns.to_flat_index()
]

print("Flattened features shape:", features_aligned.shape)
display(features_aligned.head())
```

```
Flattened features shape: (79343, 518)
```

	chroma_cens_kurtosis_01	chroma_cens_kurtosis_02	chroma_cens_kurtosis_03	chroma_cens_ku
0	-0.193837	-0.198527	0.201546	
1	-0.699535	-0.684158	0.048825	
2	-0.721487	-0.848560	0.890904	
3	-0.119708	-0.858814	2.362546	
4	-1.054053	0.932339	0.528064	

5 rows × 518 columns

In [7]:

```
# ============================================================
# 7. PREPARE X AND Y
# ============================================================

X = features_aligned.copy()
X = X.select_dtypes(include=["number"])
X = X.replace([np.inf, -np.inf], np.nan)
```

```
X = X.fillna(X.mean())
X = X.astype(np.float32)

Y = usable_df[label_cols].copy().astype(np.uint8)

print("X shape:", X.shape)
print("Y shape:", Y.shape)

print("\nAverage labels per track:", Y.sum(axis=1).mean())
print("Max labels on a track:", Y.sum(axis=1).max())
```

```
X shape: (79343, 518)
Y shape: (79343, 150)

Average labels per track: 3.67932898932483
Max labels on a track: 31
```

In [8]:

```
# ============================================================
# 8A. CREATE TRAIN / VALIDATION / TEST SPLITS
# ============================================================

split = usable_df["split"].astype(str)

train_mask = (split == "training").values
val_mask = (split == "validation").values
test_mask = (split == "test").values

X_train = X.loc[train_mask].copy()
Y_train = Y.loc[train_mask].copy()

X_val = X.loc[val_mask].copy()
Y_val = Y.loc[val_mask].copy()

X_test = X.loc[test_mask].copy()
Y_test = Y.loc[test_mask].copy()

print("Train:", X_train.shape, Y_train.shape)
print("Validation:", X_val.shape, Y_val.shape)
print("Test:", X_test.shape, Y_test.shape)
```

```
Train: (62452, 518) (62452, 150)
Validation: (8343, 518) (8343, 150)
Test: (8548, 518) (8548, 150)
```

In [9]:

```
# ============================================================
# 8B. SAMPLE THE SPLITS FOR A SAFE LAPTOP RUN
# ============================================================

SEED = 42

MAX_TRAIN_ROWS = 20000
```

```
MAX_VAL_ROWS = 5000
MAX_TEST_ROWS = 5000

if len(X_train) > MAX_TRAIN_ROWS:
    train_idx = X_train.sample(n=MAX_TRAIN_ROWS, random_state=SEED).index
    X_train = X_train.loc[train_idx].copy()
    Y_train = Y_train.loc[train_idx].copy()

if len(X_val) > MAX_VAL_ROWS:
    val_idx = X_val.sample(n=MAX_VAL_ROWS, random_state=SEED).index
    X_val = X_val.loc[val_idx].copy()
    Y_val = Y_val.loc[val_idx].copy()

if len(X_test) > MAX_TEST_ROWS:
    test_idx = X_test.sample(n=MAX_TEST_ROWS, random_state=SEED).index
    X_test = X_test.loc[test_idx].copy()
    Y_test = Y_test.loc[test_idx].copy()

print("Sampled sizes:")
print("Train:", X_train.shape, Y_train.shape)
print("Validation:", X_val.shape, Y_val.shape)
print("Test:", X_test.shape, Y_test.shape)
```

```
Sampled sizes:
Train: (20000, 518) (20000, 150)
Validation: (5000, 518) (5000, 150)
Test: (5000, 518) (5000, 150)
```

In [10]:

```
# ============================================================
# 9. SCALE FEATURES
# ============================================================

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train).astype(np.float32)
X_val_scaled = scaler.transform(X_val).astype(np.float32)
X_test_scaled = scaler.transform(X_test).astype(np.float32)

print("Scaling complete.")
```

```
Scaling complete.
```

In [11]:

```
# ============================================================
# 10. DEFINE EVALUATION AND DECODING HELPERS
# ============================================================

def evaluate_multilabel(y_true, y_pred, name="Model"):
    return {
        "Model": name,
        "Micro F1": f1_score(y_true, y_pred, average="micro", zero_division=0),
        "Macro F1": f1_score(y_true, y_pred, average="macro", zero_division=0),
```

```
        "Samples F1": f1_score(y_true, y_pred, average="samples", zero_division=0),
        "Micro Precision": precision_score(y_true, y_pred, average="micro",
zero_division=0),
        "Micro Recall": recall_score(y_true, y_pred, average="micro",
zero_division=0),
        "Hamming Loss": hamming_loss(y_true, y_pred),
        "Subset Accuracy": accuracy_score(y_true, y_pred),
        "Avg Predicted Labels": float(y_pred.sum(axis=1).mean())
    }

def get_score_matrix(model, X_input):
    if hasattr(model, "predict_proba"):
        scores = model.predict_proba(X_input)
    elif hasattr(model, "decision_function"):
        scores = model.decision_function(X_input)
    else:
        raise ValueError("Model does not support predict_proba or
decision_function.")
    return np.asarray(scores)

def decode_scalar_threshold(score_matrix, threshold, ensure_at_least_one=False):
    y_pred = (score_matrix >= threshold).astype(np.uint8)

    if ensure_at_least_one:
        row_sums = y_pred.sum(axis=1)
        zero_rows = np.where(row_sums == 0)[0]

        if len(zero_rows) > 0:
            top_idx = np.argmax(score_matrix[zero_rows], axis=1)
            y_pred[zero_rows, top_idx] = 1

    return y_pred

def decode_per_label_threshold(score_matrix, thresholds,
ensure_at_least_one=False):
    y_pred = np.zeros_like(score_matrix, dtype=np.uint8)

    for j in range(score_matrix.shape[1]):
        y_pred[:, j] = (score_matrix[:, j] >= thresholds[j]).astype(np.uint8)

    if ensure_at_least_one:
        row_sums = y_pred.sum(axis=1)
        zero_rows = np.where(row_sums == 0)[0]

        if len(zero_rows) > 0:
            top_idx = np.argmax(score_matrix[zero_rows], axis=1)
            y_pred[zero_rows, top_idx] = 1

    return y_pred

def apply_label_cap(score_matrix, y_pred, max_labels=None,
ensure_at_least_one=False):
    y_final = y_pred.copy()

    if max_labels is not None:
        for i in range(y_final.shape[0]):
            pos_idx = np.where(y_final[i] == 1)[0]
```

```
            if len(pos_idx) > max_labels:
                pos_scores = score_matrix[i, pos_idx]
                keep_order = np.argsort(-pos_scores)[:max_labels]
                keep_idx = pos_idx[keep_order]

                y_final[i, :] = 0
                y_final[i, keep_idx] = 1

    if ensure_at_least_one:
        row_sums = y_final.sum(axis=1)
        zero_rows = np.where(row_sums == 0)[0]

        if len(zero_rows) > 0:
            top_idx = np.argmax(score_matrix[zero_rows], axis=1)
            y_final[zero_rows, top_idx] = 1

    return y_final
```

In [12]:

```
# ============================================================
# 11. TRAIN THREE STRUCTURED MULTI-LABEL MODELS
# ============================================================

models = {
    "OVR SGD Logistic": OneVsRestClassifier(
        SGDClassifier(
            loss="log_loss",
            alpha=1e-4,
            max_iter=30,
            tol=1e-3,
            class_weight="balanced",
            random_state=42
        ),
        n_jobs=1
    ),
    "OVR SGD Hinge": OneVsRestClassifier(
        SGDClassifier(
            loss="hinge",
            alpha=1e-4,
            max_iter=30,
            tol=1e-3,
            class_weight="balanced",
            random_state=42
        ),
        n_jobs=1
    ),
    "OVR Passive Aggressive": OneVsRestClassifier(
        PassiveAggressiveClassifier(
            C=1.0,
            max_iter=30,
            tol=1e-3,
            class_weight="balanced",
            random_state=42
```

```
        ),
        n_jobs=1
    )
}

trained_models = {}
val_score_dict = {}
validation_model_results = []

for model_name, model in models.items():
    print(f"\nTraining {model_name} ...")
    model.fit(X_train_scaled, Y_train)

    trained_models[model_name] = model
    val_scores = get_score_matrix(model, X_val_scaled)
    val_score_dict[model_name] = val_scores

    # Default threshold
    if "Logistic" in model_name:
        default_threshold = 0.5
    else:
        default_threshold = 0.0

    Y_val_pred = decode_scalar_threshold(
        val_scores,
        threshold=default_threshold,
        ensure_at_least_one=False
    )

    results = evaluate_multilabel(Y_val, Y_val_pred, name=model_name)
    results["Default Threshold"] = default_threshold

    validation_model_results.append(results)
    print(results)

validation_models_df = pd.DataFrame(validation_model_results).sort_values(
    "Micro F1", ascending=False
).reset_index(drop=True)

print("\nValidation model comparison:")
display(validation_models_df)
```

```
Training OVR SGD Logistic ...
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
```

```
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
```

```
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
```

```
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
```

```
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
```

```
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
```

```
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
{'Model': 'OVR SGD Logistic', 'Micro F1': 0.16195028974591513, 'Macro F1':
0.08093826711139597, 'Samples F1': 0.16160643833136884, 'Micro Precision':
0.095269958676565, 'Micro Recall': 0.5396711570967577, 'Hamming Loss':
0.14538933333333334, 'Subset Accuracy': 0.0, 'Avg Predicted Labels': 22.1182,
'Default Threshold': 0.5}

Training OVR SGD Hinge ...
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
```

```
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
```

```
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
```

```
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
```

```
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
```

```
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
```

```
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
{'Model': 'OVR SGD Hinge', 'Micro F1': 0.16533077749522712, 'Macro F1':
0.08169435589256198, 'Samples F1': 0.16511085222550514, 'Micro Precision':
0.09735649134305532, 'Micro Recall': 0.5478153972237874, 'Hamming Loss':
0.14398266666666668, 'Subset Accuracy': 0.0, 'Avg Predicted Labels': 21.9708,
'Default Threshold': 0.0}

Training OVR Passive Aggressive ...
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-
packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
Maximum number of iteration reached before convergence. Consider increasing max_iter
to improve the fit.
  warnings.warn(
```

```
{'Model': 'OVR Passive Aggressive', 'Micro F1': 0.12076803997718694, 'Macro F1':
0.0688227844316684, 'Samples F1': 0.12008075739428518, 'Micro Precision':
0.06754687632912469, 'Micro Recall': 0.5694309276238283, 'Hamming Loss': 0.215828,
'Subset Accuracy': 0.0, 'Avg Predicted Labels': 32.9164, 'Default Threshold': 0.0}

Validation model comparison:
```

	Model	Micro F1	Macro F1	Samples F1	Micro Precision	Micro Recall	Hamming Loss	Subset Accuracy	Av Predicte Labe
0	OVR SGD Hinge	0.165331	0.081694	0.165111	0.097356	0.547815	0.143983	0.0	21.970
1	OVR SGD Logistic	0.161950	0.080938	0.161606	0.095270	0.539671	0.145389	0.0	22.118
2	OVR Passive Aggressive	0.120768	0.068823	0.120081	0.067547	0.569431	0.215828	0.0	32.916

In [13]:

```
# ============================================================
# 12. SELECT BEST BASE MODEL
# ============================================================
```

```
best_model_name = validation_models_df.iloc[0]["Model"]
best_model = trained_models[best_model_name]

val_scores_best = val_score_dict[best_model_name]
test_scores_best = get_score_matrix(best_model, X_test_scaled)

print("Best base model:", best_model_name)
print("Validation score matrix shape:", val_scores_best.shape)
print("Test score matrix shape:", test_scores_best.shape)
```

```
Best base model: OVR SGD Hinge
Validation score matrix shape: (5000, 150)
Test score matrix shape: (5000, 150)
```

In [14]:

```
# ============================================================
# 13. LEARN PER-LABEL THRESHOLDS FOR THE BEST MODEL
# ============================================================

if "Logistic" in best_model_name:
    per_label_grid = np.array([0.1, 0.3, 0.5, 0.7, 0.9])
else:
    per_label_grid = np.array([-1.0, -0.5, 0.0, 0.5, 1.0])

best_thresholds_per_label = np.zeros(val_scores_best.shape[1], dtype=np.float32)

for j in range(val_scores_best.shape[1]):
    y_true_j = Y_val.iloc[:, j].values
    scores_j = val_scores_best[:, j]

    best_f1_j = -1.0
    best_t_j = per_label_grid[0]

    for t in per_label_grid:
        y_pred_j = (scores_j >= t).astype(np.uint8)
        f1_j = f1_score(y_true_j, y_pred_j, zero_division=0)

        if f1_j > best_f1_j:
            best_f1_j = f1_j
            best_t_j = t

    best_thresholds_per_label[j] = best_t_j

print("Per-label thresholds learned.")
print("Unique thresholds selected:", np.unique(best_thresholds_per_label))
```

```
Per-label thresholds learned.
Unique thresholds selected: [-1.  -0.5  0.   0.5  1. ]
```

In [15]:

```
# ============================================================
# 14. VALIDATION SEARCH: LABEL-CAP DECODING
```

```
# ============================================================

base_val_pred = decode_per_label_threshold(
    val_scores_best,
    best_thresholds_per_label,
    ensure_at_least_one=True
)

cap_results = []

for cap in [None, 15, 12, 10, 8, 6, 5, 4, 3]:
    Y_val_pred_cap = apply_label_cap(
        val_scores_best,
        base_val_pred,
        max_labels=cap,
        ensure_at_least_one=True
    )

    cap_name = "Per-Label Thresholds" if cap is None else f"Per-Label Thresholds +
Cap-{cap}"

    results = evaluate_multilabel(Y_val, Y_val_pred_cap, name=cap_name)
    results["Cap"] = "None" if cap is None else cap

    cap_results.append(results)

cap_results_df = pd.DataFrame(cap_results).sort_values(
    "Micro F1", ascending=False
).reset_index(drop=True)

print("Validation label-cap tuning results:")
display(cap_results_df)
```

```
Validation label-cap tuning results:
```

	Model	Micro F1	Macro F1	Samples F1	Micro Precision	Micro Recall	Hamming Loss	Subset Accuracy	Av Predicte Labe
0	Per-Label Thresholds	0.168899	0.082488	0.168700	0.099519	0.557701	0.142871	0.0	21.881
1	Per-Label Thresholds + Cap-15	0.112167	0.074160	0.110382	0.070851	0.269067	0.110877	0.0	14.828
2	Per-Label Thresholds + Cap-12	0.086872	0.067720	0.084694	0.057603	0.176612	0.096647	0.0	11.971
3	Per-Label Thresholds + Cap-10	0.070943	0.062179	0.068724	0.049331	0.126261	0.086083	0.0	9.993
4	Per-Label Thresholds + Cap-8	0.054774	0.054661	0.052625	0.040756	0.083491	0.075009	0.0	7.998
5	Per-Label Thresholds + Cap-6	0.039296	0.045746	0.037442	0.032435	0.049839	0.063433	0.0	5.999
6	Per-Label Thresholds + Cap-5	0.032972	0.041579	0.031237	0.029360	0.037597	0.057407	0.0	5.000
7	Per-Label Thresholds + Cap-4	0.025555	0.036157	0.024204	0.025250	0.025867	0.051351	0.0	4.000
8	Per-Label Thresholds + Cap-3	0.019060	0.031471	0.017810	0.021933	0.016852	0.045153	0.0	3.000

In [16]:

```
# ============================================================
# 15. SELECT BEST DECODING RULE
# ============================================================

best_cap_row = cap_results_df.iloc[0].to_dict()
best_cap_value = best_cap_row["Cap"]

print("Best decoding rule on validation:")
print(best_cap_row)
```

```
Best decoding rule on validation:
{'Model': 'Per-Label Thresholds', 'Micro F1': 0.16889916155403362, 'Macro F1':
0.08248818923043892, 'Samples F1': 0.16869968682142966, 'Micro Precision':
0.09951922198051295, 'Micro Recall': 0.5577011729754648, 'Hamming Loss':
0.14287066666666667, 'Subset Accuracy': 0.0, 'Avg Predicted Labels': 21.8812, 'Cap':
'None'}
```

In [17]:

```python
# ============================================================
# 16. APPLY BEST DECODING RULE TO TEST
# ============================================================

base_test_pred = decode_per_label_threshold(
    test_scores_best,
    best_thresholds_per_label,
    ensure_at_least_one=True
)

if best_cap_value == "None":
    final_test_pred = base_test_pred.copy()
else:
    final_test_pred = apply_label_cap(
        test_scores_best,
        base_test_pred,
        max_labels=int(best_cap_value),
        ensure_at_least_one=True
    )

final_test_results = evaluate_multilabel(
    Y_test,
    final_test_pred,
    name=f"{best_model_name} + Best Decoding"
)

print("Final structured test results:")
print(final_test_results)
```

```
Final structured test results:
{'Model': 'OVR SGD Hinge + Best Decoding', 'Micro F1': 0.16342705578617783, 'Macro
F1': 0.08431567849564779, 'Samples F1': 0.16373645996770694, 'Micro Precision':
0.0955557798720784, 'Micro Recall': 0.5640845070422535, 'Hamming Loss':
0.14214266666666667, 'Subset Accuracy': 0.0, 'Avg Predicted Labels': 21.7946}
```

In [18]:

```python
# ============================================================
# 17. BUILD TEST COMPARISON TABLE
# ============================================================

# Default test result for the best base model
if "Logistic" in best_model_name:
    default_threshold = 0.5
else:
    default_threshold = 0.0
```

```
Y_test_pred_default = decode_scalar_threshold(
    test_scores_best,
    threshold=default_threshold,
    ensure_at_least_one=False
)

default_test_results = evaluate_multilabel(
    Y_test,
    Y_test_pred_default,
    name=f"{best_model_name} - Default"
)

per_label_only_test_pred = decode_per_label_threshold(
    test_scores_best,
    best_thresholds_per_label,
    ensure_at_least_one=True
)

per_label_only_test_results = evaluate_multilabel(
    Y_test,
    per_label_only_test_pred,
    name=f"{best_model_name} - Per-Label Only"
)

test_comparison_df = pd.DataFrame([
    default_test_results,
    per_label_only_test_results,
    final_test_results
])

print("Structured test comparison:")
display(test_comparison_df)
```

```
Structured test comparison:
```

	Model	Micro F1	Macro F1	Samples F1	Micro Precision	Micro Recall	Hamming Loss	Subset Accuracy	Avg Predicted Labels
0	OVR SGD Hinge - Default	0.160855	0.083810	0.161082	0.093984	0.557584	0.143191	0.0	21.9038
1	OVR SGD Hinge - Per-Label Only	0.163427	0.084316	0.163736	0.095556	0.564085	0.142143	0.0	21.7946
2	OVR SGD Hinge + Best Decoding	0.163427	0.084316	0.163736	0.095556	0.564085	0.142143	0.0	21.7946

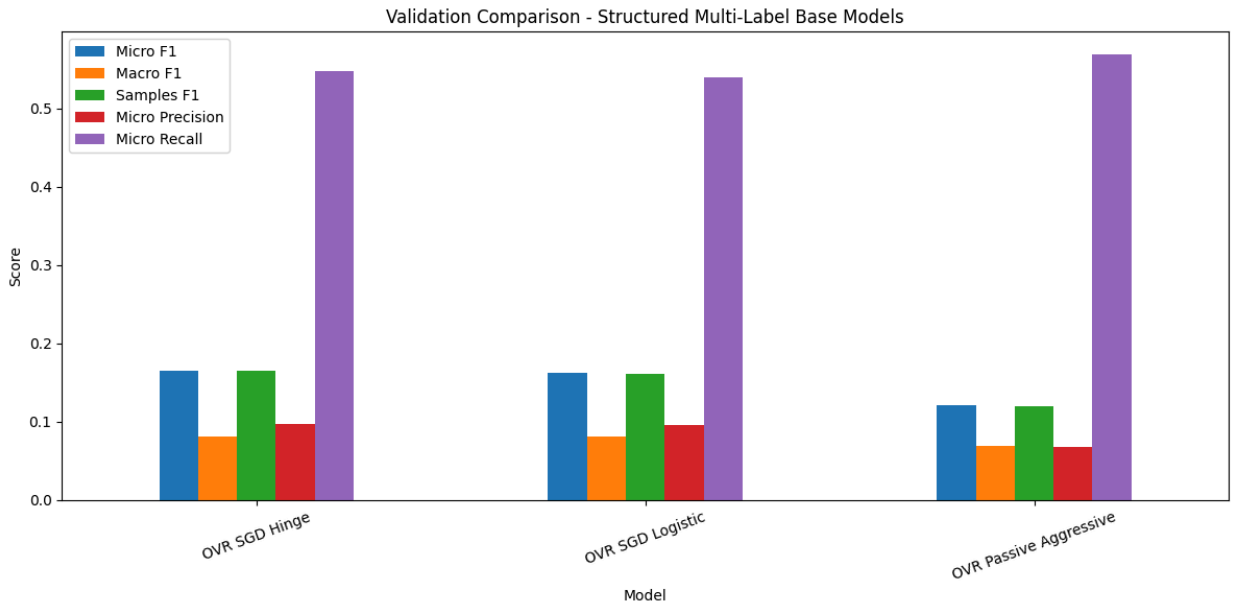
In [19]:

```
# ============================================================
# 18. PLOT VALIDATION MODEL COMPARISON
# ============================================================

plot_cols = ["Micro F1", "Macro F1", "Samples F1", "Micro Precision", "Micro
Recall"]

plot_df = validation_models_df.set_index("Model")[plot_cols]

ax = plot_df.plot(kind="bar", figsize=(12, 6))
ax.set_title("Validation Comparison - Structured Multi-Label Base Models")
ax.set_ylabel("Score")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

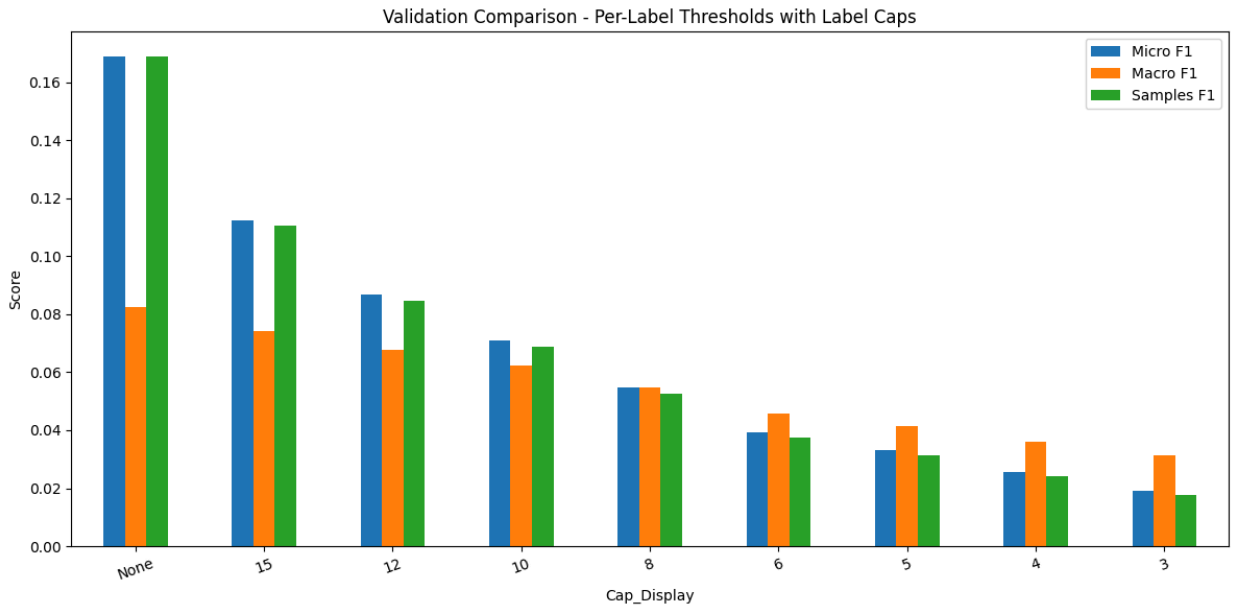


In [20]:

```
# ============================================================
# 19. PLOT LABEL-CAP VALIDATION SEARCH
# ============================================================

plot_cap_df = cap_results_df.copy()
plot_cap_df["Cap_Display"] = plot_cap_df["Cap"].astype(str)

ax = plot_cap_df.set_index("Cap_Display")[["Micro F1", "Macro F1", "Samples
F1"]].plot(
    kind="bar",
    figsize=(12, 6)
)
ax.set_title("Validation Comparison - Per-Label Thresholds with Label Caps")
ax.set_ylabel("Score")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [21]:

```
# ============================================================
# 20. SAVE OUTPUTS
# ============================================================

os.makedirs("../models", exist_ok=True)
os.makedirs("../data/processed", exist_ok=True)

joblib.dump(best_model,
"../models/final_structured_multilabel_candidate150_best_model.joblib")
joblib.dump(scaler,
"../models/final_structured_multilabel_candidate150_scaler.joblib")

validation_models_df.to_csv(

"../data/processed/final_structured_multilabel_candidate150_model_comparison.csv",
    index=False
)

cap_results_df.to_csv(

"../data/processed/final_structured_multilabel_candidate150_cap_tuning_results.csv"
,
    index=False
)

test_comparison_df.to_csv(

"../data/processed/final_structured_multilabel_candidate150_test_comparison.csv",
    index=False
)

pd.DataFrame([final_test_results]).to_csv(
    "../data/processed/final_structured_multilabel_candidate150_test_results.csv",
    index=False
)
```

```
np.save(
    "../data/processed/final_structured_multilabel_candidate150_y_test_pred.npy",
    final_test_pred
)

np.save(

"../data/processed/final_structured_multilabel_candidate150_best_thresholds_per_lab
el.npy",
    best_thresholds_per_label
)

with
open("../data/processed/final_structured_multilabel_candidate150_best_model_name.tx
t", "w") as f:
    f.write(best_model_name)

with
open("../data/processed/final_structured_multilabel_candidate150_best_cap.txt",
"w") as f:
    f.write(str(best_cap_value))

print("Saved final structured multi-label improvement outputs.")
```

```
Saved final structured multi-label improvement outputs.
```

In [22]:

```
# ============================================================
# 21. INTERPRETATION NOTES
# ============================================================

print("1. This notebook performs one final structured upgrade pass for the
candidate-150 multi-label task.")
print("2. Three structured base models were compared: OVR SGD Logistic, OVR SGD
Hinge, and OVR Passive Aggressive.")
print("3. Per-label thresholds were learned for the best model using validation
data.")
print("4. A label-cap decoding rule was tuned to reduce over-prediction.")
print("5. The saved result becomes the final improved structured baseline before
the audio multi-label phase.")
```

```
1. This notebook performs one final structured upgrade pass for the candidate-150
multi-label task.
2. Three structured base models were compared: OVR SGD Logistic, OVR SGD Hinge, and
OVR Passive Aggressive.
3. Per-label thresholds were learned for the best model using validation data.
4. A label-cap decoding rule was tuned to reduce over-prediction.
5. The saved result becomes the final improved structured baseline before the audio
multi-label phase.
```